

Research Proposal

Can Reinforcement Learning Improve Budget-Constrained Recovery in Networks Subject to Cascading Failures?

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Abstract

This proposal studies the recovery of graph-structured networks subject to cascading failures and limited patching capacity. The central question is whether reinforcement learning can learn recovery policies that restore network functionality more effectively than standard healing heuristics under limited capacity. The project is motivated by a gap between three related works that have largely developed in parallel: graph reinforcement learning for node-level intervention, sequential decision making on networks under constraints, and resilience-oriented recovery under cascading failure. Existing work shows that reinforcement learning can learn transferable intervention policies on graphs, that constraints can be embedded into sequential decision problems, and that cascading-failure mitigation can be studied through recovery or control strategies. However, little work combines these elements into a single framework for learning constrained recovery policies that generalize across network topologies.

Methodologically, the project models cascading failure through a load-capacity framework on graph networks and formulates sequential node reactivation as a Markov Decision Process. A graph-based reinforcement learning agent is trained on synthetic networks, primarily small Barabási–Albert graphs, and evaluated against standard recovery heuristics on both (augmented) synthetic and real-world network topologies. The main contribution of the project is twofold: first, it extends graph reinforcement learning from network dismantling to network recovery under cascading disruption; second, it uses the learned policy to estimate the minimum recovery capacity required to achieve a predefined resilience target. The project thereby contributes a tractable framework for studying resilience-aware recovery and offers a more policy-relevant output than static node rankings alone.

1 Academic and Practical Relevance

1.1 Academic Relevance

This project is academically relevant because it connects three active research areas that are rarely combined in a single framework: graph reinforcement learning, budget-constrained sequential decision making, and network resilience under cascading failure. Existing graph-RL research has mainly focused on node removal, influence maximization, or cyber defense, while resilience research often relies on heuristics or optimization-based restoration methods. By studying budget-constrained

recovery in disrupted networks, this project extends graph reinforcement learning to a new class of resilience problems in which actions must be allocated sequentially under uncertainty.

The project is also relevant for machine learning research more specifically. It studies how graph-based reinforcement learning can operate in environments with nonlinear cascade dynamics, and generalization across network topologies. In doing so, it contributes to the broader question of whether learned graph policies can outperform standard handcrafted heuristics in structured combinatorial decision problems.

1.2 Practical Relevance

The project is practically relevant for infrastructure operators, resilience planners, and policy makers concerned with the robustness of networked systems. In real recovery planning, it is not enough to know which nodes are important in a static sense. Decision makers need guidance on how limited recovery resources should be allocated over time and how much intervention capacity is required to maintain system functionality under disruption.

By estimating the minimum recovery budget needed to achieve a predefined resilience target, the project aims to provide a more policy-relevant output than simple node rankings alone. Even in a stylized setting, such threshold estimates can help inform planning decisions about redundancy, recovery capacity, and the prioritization of interventions in infrastructure-like systems exposed to cascading failure.

2 Literature Review and Positioning

This proposal lies at the intersection of graph reinforcement learning, budget-constrained sequential decision making, and network resilience under cascading failure. The central research question is whether reinforcement learning can improve budgeted recovery in networks subject to cascading failures. The relevant literature therefore includes not only graph-based RL for node-level intervention, but also work on cost-aware resource allocation, cyber defense, and resilience-oriented recovery in disrupted networked systems.

2.1 Graph Reinforcement Learning for Network Intervention

The main methodological starting point is FINDER, which formulates node intervention in complex networks as a sequential decision problem and learns a policy on small synthetic graphs that generalizes to larger unseen networks Fan et al. (2020). FINDER is the closest methodological anchor for this proposal because it shows that graph intervention can be learned through reinforcement learning rather than relying exclusively on handcrafted centrality heuristics or combinatorial optimization. More broadly, recent surveys confirm that graph reinforcement learning has become a promising framework for combinatorial decision problems on structured networks Darvariu et al. (2024), including critical-node identification Chen (2025).

More recent work has refined this intervention logic in the network dismantling setting. MaxShot improves the dismantling objective by combining graph representation learning with reinforcement learning and explicitly prioritizing high-degree nodes within the giant connected component Liu et al. (2024). GRLND further advances graph reinforcement learning for dismantling by learning

node-removal strategies through a graph-aware policy architecture that accounts for joint intervention effects Qu et al. (2025). These studies strengthen the methodological case that graph-based RL can learn effective intervention rules on complex networks and transfer them across graph instances.

This proposal follows that line of work in two key respects. First, it adopts the sequential graph-intervention framing of FINDER and the broader ambition of cross-graph generalization. Second, it uses graph-aware state representations to evaluate interventions under dynamically changing network conditions. However, it also departs from the dismantling literature in an important way: rather than selecting nodes to remove in order to fragment a network, the agent selects failed nodes to reactivate in order to preserve or restore network functionality under cascading disruption.

2.2 Budget-Constrained Sequential Intervention

A second relevant strand of the literature studies intervention under explicit resource constraints. In graph-based influence maximization, residual graph reinforcement learning has been used to model finite intervention resources directly within the state and action design, allowing the learned policy to reason over budget usage dynamically Ou et al. (2025). Similarly, adaptive budgeted influence maximization has been formulated as a sequential reinforcement learning problem with intervention costs and finite budgets Fadda et al. (2024). At a more general level, budgeted reinforcement learning provides a broader conceptual foundation for learning under resource constraints Carrara et al. (2019).

Although these studies are not concerned with network recovery or resilience, they are highly relevant for the present proposal because they show how limited resources can be incorporated into reinforcement learning on graphs as part of the decision process itself. This proposal builds on that logic by treating recovery budget as a state variable that constrains which recovery actions remain feasible over time. The key difference is that the objective is not diffusion or influence maximization, but resilience preservation under disruption.

2.3 Graph-Based Reinforcement Learning for Defense

A third body of work moves closer to the application domain by studying defensive decision making on graph-structured cyber environments. Nyberg and Johnson show that reinforcement learning can learn automated defense strategies in graph-based cyber attack simulations and outperform heuristic baselines while accounting for explicit action costs Nyberg and Johnson (2023). Their later work further demonstrates that message-passing neural networks combined with reinforcement learning can support structural generalization across changed network topologies, which is particularly relevant for learning policies on synthetic graphs and evaluating them on unseen real-world systems Nyberg and Johnson (2024). Related work by Le et al. (2024). integrates reinforcement learning with attack-graph risk modeling and explicitly incorporates residual risk, control efficacy, and cost into the reward structure Le et al. (2024). Recent work also continues to push toward more generalizable graph-based cyber defense agents King et al. (2025), while hierarchical multi-agent RL has been explored for cyber network defense in larger-scale settings Singh et al. (2024).

These papers are important because they establish that graph-based reinforcement learning is not limited to offensive or dismantling tasks, but can also support defense-oriented decision making under uncertainty. However, most of this literature focuses on cyber attack graphs and attacker–

defender environments rather than on cascading-failure dynamics in infrastructure-like networks.

2.4 Cascading Failure, Recovery, and Resilience

The final strand of literature concerns cascading failure and resilience in networked systems. Foundational work models cascading failures as load redistribution processes on complex networks and studies how network topology and redistribution rules affect systemic collapse (Wang and Rong, 2009; Wu et al., 2008; Liu et al., 2015). More recent studies extend this literature by considering adjustable load and capacity parameters (Feng et al., 2023), as well as concurrent self-healing and resilience trade-offs under finite budgets (Al-Aqqad et al., 2022; Al Aqqad, 2025). This line of work provides the core motivation for representing failures, overload, and recovery within a load-capacity framework.

More directly relevant are recent resilience-oriented recovery studies. Song et al. (2025) propose a sequential recovery strategy for complex networks under cascading failure and explicitly analyze recovery ordering under practical constraints, making their work especially useful for motivating benchmark healing heuristics in the present project. In related infrastructure settings, deep reinforcement learning has been used to mitigate cascading failure in power grids through sequential remedial actions (Zhang et al., 2024b), and multi-stage cascading failure mitigation has also been studied using deep RL (Meng et al., 2025). Beyond RL, optimization-based work has examined the joint design of reliability improvement and resilience under mixed cascading failures in dependent networks (Zhou et al., 2023), while restoration of infrastructure networks under limited repair resources has also been approached using deep reinforcement learning (Feng et al., 2025).

Together, these studies show that cascading-failure mitigation and recovery are naturally sequential, resource-constrained decision problems. However, the existing literature still leaves a gap. Recovery under cascading failure is often handled through heuristics, optimization, or domain-specific control strategies, while graph reinforcement learning for transferable node-level intervention has mainly been developed in dismantling, interdiction, or cyber defense contexts (Zhang et al., 2024a; Jiang et al., 2025). Few studies combine all three ingredients central to this proposal: graph-based policy learning, explicit budget-constrained recovery decisions, and resilience evaluation through a cascading-failure model that supports minimum-budget estimation.

2.5 Overall Positioning and Contribution

The papers closest to the proposed research are therefore FINDER (Fan et al., 2020), which provides the main methodological template for transferable graph-based intervention learning, and recent defense and resilience studies such as Nyberg and Johnson (2023); Le et al. (2024); Zhang et al. (2024b), which show that reinforcement learning can support cost-aware recovery or defense under disruption. The proposed project combines these perspectives but changes the problem structure in a meaningful way.

This study proposes a graph-based reinforcement learning framework for budget-constrained recovery in networks exposed to cascading failures. Building on the intervention logic of FINDER, it extends graph RL from network dismantling to network protection and restoration. Its first contribution is the development of a graph-based recovery policy that sequentially allocates limited

recovery resources under uncertainty. Its second contribution is the use of the learned policy to estimate the minimum recovery budget required to satisfy a predefined resilience target. The project therefore contributes a tractable extension of graph reinforcement learning to budget-constrained recovery and offers an interpretable resource-based measure of resilience in disrupted networked systems.

3 Data

Following prior work on graph-based reinforcement learning, we train the model on synthetic networks to ensure scalability and generalization. In particular, we generate graphs with 30-50 nodes each, with a focus on Barabási–Albert networks, which capture the heavy-tailed degree distributions commonly observed in real-world systems (Fan et al., 2020). Training on small to medium-sized synthetic graphs allows efficient learning while exposing the agent to diverse structural patterns. For testing, we use both synthetic graphs with real-world network datasets in which cascading failure and recovery are commonly studied, such as power transmission networks, transportation systems, and interdependent infrastructure networks Zhang et al. (2024b); Song et al. (2025); Zhou et al. (2023). These examples are well aligned with the resilience literature because they represent networked systems in which local disruption can propagate and recovery decisions must be prioritized under limited resources. All graphs are augmented with load and capacity following the model in Feng et al. (2023). This setup enables us to assess whether the learned policy transfers to realistic network structures and maintains performance under more complex and heterogeneous conditions.

4 Methodology

4.1 Graph features and Cascading failure

To evaluate the performance of reinforcement learning on network resilience with stochastic initial node failures, we combine a cascading-failure framework with a reinforcement learning recovery policy that learns which failed nodes to reactivate under a limited recovery budget. We use a graph $G = (V, E)$, with V being the set of nodes and E being the set of edges. The edges are unweighted and undirected and hence purely indicate connectivity between two nodes. Each node i has a maximum capacity C_i and a current load $L_i^{(t)}$

We model initial loads and node capacities, as well as load redistribution cascade following Feng et al. (2023). Initial loads $L_i^{(0)}$ are modeled as a function of the node’s degree. While the exact function varies based on application, the model is well documented in literature, motivated by the real-life properties of networks, where highly connected nodes often handle more traffic.

In similar manner, nodes with higher traffic are often built with higher capacity, leading to the model proposed by (Motter and Lai, 2003), where $C_i = (1 + \alpha)L_i^{(0)}$, with α being a variable tolerance parameter. As a baseline we set $\alpha = 0.2$, representing a moderate spare capacity regime. We then vary α in sensitivity analysis, because the cascading failure literature shows that the tolerance parameter materially affects resilience outcomes and should therefore not be treated as universally fixed.

We choose the failure to happen at random in time $t = 0$, by uniformly choosing nodes out of the set of active nodes. Following Motter and Lai (2002), we assign each node an equal probability p_{fail} of failing. As higher values of p_{fail} initiate more node failures, we will test different values to evaluate the impact of this parameter on our results.

Each failed node redistributes its load to its active neighboring nodes. The load redistribution that starts the cascade failure follows Feng et al. (2023). Suppose node i^* fails at time t . Then the load $L_{i^*}^{(t)}$ is redistributed to its neighboring nodes. Define $S(i^*)$ as the set of active nodes adjacent to node i^* . Then each of the neighboring nodes obtains a share of the load, proportional to their capacity relative to the total capacity of the nodes in $S(i^*)$:

$$\Delta L_j^{(t)} = L_{i^*}^{(t)} \frac{C_j}{\sum_{m \in S(i^*)} C_m}, \forall j \in S(i^*). \quad (1)$$

This mirrors the Local Weighted Flow Redistribution Rule (LWFRR), where higher-capacity nodes that are “equipped” to absorb extra flow better than lower-capacity nodes, reducing the risks of “overflowing” or propagating cascades (Wu et al., 2008). Then, the load of the neighboring nodes is updated by

$$L_j^{(t+1)} = \Delta L_j^{(t)} + L_j^{(t)}. \quad (2)$$

Each node j whose updated load exceeds its capacity $L_j^{(t+1)} > C_j$, fails in the subsequent round. This cascade is repeated until no further nodes fail.

4.2 Sequential Recovery with Reinforcement Learning

We model network resilience by introducing a node-reactivation mechanism. That means that in each time step, a fixed number B of nodes can be reactivated. Reactivated nodes are interpreted as repaired or recovered nodes that re-enter the system with full available capacity and no immediate assigned burden, so that they can absorb future redistributed load. Reactivation assigns the node its original capacity but a load of zero. This way, the cascade can be interrupted or even entirely stopped, depending on the number of reactivations in each round.

Reinforcement learning is well suited to the self-healing problem because recovery under cascading failure is inherently a sequential decision problem under uncertainty. After each cascade, the agent observes the updated graph, including which nodes failed and how connectivity changed, and must choose recovery actions whose value depends not only on the current state but also on their future effects.

We therefore adopt FINDER (Fan et al., 2020) as the methodological starting point, because it was specifically designed to learn graph-based intervention policies through sequential node selection and has been shown to generalize well across graph topologies and sizes. In our setting, however, the goal is reversed: instead of removing nodes to reduce connectivity, the agent reactivates failed nodes to restore connectivity. Moreover, our constraint implies that the agent may reactivate B nodes in one round. Rather than treating this as a one shot combinatorial action over all subsets of failed nodes, which would make the action space intractably large, we follow the sequential logic of FINDER and model each round as B consecutive reactivation decisions.

The budget is therefore incorporated as a state variable and feasibility constraint. This pre-

serves the main advantages of FINDER, namely node level scoring, masking of invalid actions, and transferability across graphs, while adapting the reward function so that higher post recovery connectivity is preferred rather than lower connectivity as in the original dismantling problem (Fan et al., 2020; Ou et al., 2025; Meng et al., 2025).

We retain the FINDER connectivity based objective, but reverse its direction: instead of rewarding actions that reduce accumulated normalized connectivity, we reward actions that increase post cascade connectivity. Since our RL sequentially reactivates failed nodes, the reward is naturally defined as the marginal increase in ANC after each reactivation step.

4.3 Minimum Recovery Budget

A central object of interest is the minimum recovery budget required to keep the network above a predefined resilience target. Examining this threshold is substantively important because, in infrastructure settings, operators typically do not only want to know whether recovery is possible, but also how many interventions per round are needed to prevent persistent cascade propagation or restore connectivity within an acceptable time horizon. Recent resilience studies therefore treat budget as a key control parameter rather than a purely technical detail. In particular, concurrent self-healing models show that resiliency is highly sensitive to budget size and may exhibit sharp threshold effects, where small changes in available recovery resources can move the system from non-recovery to rapid recovery (Al Aqqad, 2025). Likewise, sequential recovery models for complex networks explicitly incorporate resource limits and upper bounds on simultaneous recovery actions, showing that recovery capacity materially shapes resilience outcomes. More broadly, restoration research in infrastructure networks models recovery as a resource-constrained optimization problem in which the allocation of limited repair capacity determines both restoration speed and final system performance (Feng et al., 2025).

In our framework, the budget parameter B determines the number of failed nodes that can be reactivated in each round. Increasing B enlarges the feasible action space and allows the agent to construct larger recovery sets, which should weakly improve the best attainable resilience outcome. For this reason, the optimal value function is expected to be weakly monotone in B , even if empirical estimates from a learned policy are not perfectly monotone because of finite training, cascades following initial random failure, or approximation error. This monotonicity intuition is consistent with the broader literature on budget-constrained graph optimization, where larger budgets expand the set of admissible interventions and typically improve the achievable objective value Ou et al. (2025). We therefore estimate the minimum budget

$$B^* = \min\{B : \mathbb{E}[\text{ANC}(G_T)] \geq \tau|\pi\}, \quad (3)$$

where τ is a predefined resilience threshold and π is the learned policy or applied benchmark heuristic. The expectation is taken over repeated stochastic cascade and recovery simulations. This converts the recovery problem into a policy-relevant threshold question: how many node reactivations per round are sufficient for the learned policy to maintain network functionality above the target level.

4.4 Reinforcement Learning Formulation

We formalize the recovery problem as a Markov Decision Process (MDP), in which a reinforcement learning agent sequentially reactivates failed nodes under a budget constraint to improve network resilience.

State. At time step t , the state s_t consists of:

- the current graph structure $G_t = (V_t, E_t)$,
- the load of each node $\{L_i^{(t)}\}_{i \in V_t}$,
- the set of failed nodes $F_t \subseteq V_t$,
- the recovery budget B_t .

The state therefore captures both the physical condition of the network and the remaining decision capacity of the agent.

Action. At each decision step, the agent selects one failed node to reactivate:

$$a_t \in F_t.$$

Reactivating node i resets its load to zero and restores its capacity, allowing it to participate again in load redistribution. Actions are taken sequentially until the budget B is exhausted.

Transition. After each action, the environment evolves according to two mechanisms:

1. **Reactivation update:** The selected node is restored to the active set with zero load.
2. **Cascade dynamics:** Load redistribution may trigger further node failures if capacity constraints are exceeded, following the cascading failure model described in subsection 4.1.

Reward. The reward is defined as the marginal improvement in network connectivity after each reactivation step. Let $\text{ANC}(G_t)$ denote the accumulated normalized connectivity at time t . Then the reward is given by:

$$r_t = \text{ANC}(G_t) - \text{ANC}(G_{t-1}).$$

ANC is used as the main operational measure of network functionality, so the reward treats marginal connectivity restoration as the immediate objective of the recovery policy. This encourages the agent to select nodes whose reactivation most effectively restores connectivity and mitigates cascading failures. Even if not directly optimized, recovery time and threshold attainment are evaluated separately.

Objective. The agent aims to learn a policy $\pi(a_t | s_t)$ that maximizes the expected cumulative reward over an episode:

$$\mathbb{E} \left[\sum_{t=0}^T r_t \right],$$

where T is the time horizon determined by the cascade and recovery process. This corresponds to maximizing network resilience under a constrained recovery budget.

4.5 Evaluation Design

We compare the learned RL recovery policy with a set of non-learning recovery heuristics under identical initial failure and subsequent cascade environments and identical per-round recovery budgets. Each policy is evaluated over repeated Monte Carlo simulations across multiple tolerance levels α .

The first baseline is a random recovery policy, which selects B failed nodes uniformly at random for reactivation at each step. This serves as a lower-bound reference, capturing the expected performance of a policy with no structural knowledge of the network.

The second and third baselines exploit local structural information. The highest-degree-first policy reactivates failed nodes in descending order of their degree, prioritizing nodes that were originally the most connected and therefore likely the most critical to network throughput. The highest-overload-risk policy instead focuses on neighborhood stress, reactivating failed nodes whose active neighbors are currently operating closest to their capacity limits, thereby targeting nodes whose absence most directly threatens further cascade propagation. Another proxy could be the betweenness-centrality-first recovery, with nodes that serve as critical bridges in the network’s structure being prioritized in the recovery process.

Finally, we evaluate the greedy one-step connectivity gain policy, which selects the failed node whose reactivation yields the largest immediate increase of ANC at each step. Unlike the other heuristics, this policy explicitly optimizes a global structural objective at each step, making it the strongest and most computationally expensive non-learning baseline.

To measure efficiency we measure different outcome metrics across all methods. These include:

1. The final ANC at horizon T
2. The minimum budget B^*

$$B^* = \min\{B : \mathbb{E}[\text{ANC}(G_T)] \geq \tau|\pi\}$$

3. Recovery time

Number of rounds needed to return above a target resilience threshold

To assess generalization, the evaluation varies systematically across network topologies as well as across budgets B and cascade parameters. In particular, we evaluate the impact on our results of varying the capacity-parameter α and the initial failure probability p_{fail} .

To ensure comparability, we use the same test graphs for all methods. Where possible, we use the same random seeds for matched failure realizations, to minimize the influence of randomness on our results. Furthermore, we report confidence intervals for all outcomes instead of precise values.

5 Expected Findings and Contribution

The working expectation is that a graph-based reinforcement learning recovery policy will outperform standard healing heuristics when cascading failures generate nonlinear and path-dependent disruptions. In particular, we expect the learned policy to make better use of limited recovery capacity than degree-based, load-based, or other myopic recovery rules, because it can account for both the current network state and the future effects of reactivating specific failed nodes.

The contribution of the project is not to claim a universally optimal recovery strategy, but to provide a framework for learning and evaluating constrained recovery policies under cascading failure. In academic terms, the project contributes to graph reinforcement learning, network resilience, and sequential node selection under cascading failure. In practical terms, it offers a more policy-relevant answer than static node rankings or simple restoration heuristics, because it returns both an actionable recovery policy and an estimate of the minimum recovery budget required to achieve a predefined resilience target.

6 Feasibility and Limitations

The project is feasible for a seminar for two main reasons. First, it relies on a stylized load-capacity cascade model on synthetic and real-world network topologies rather than requiring detailed calibration of a specific infrastructure system. This keeps the environment rich enough to study cascading failure and sequential recovery, while remaining computationally manageable. Second, the project focuses on a single recovery agent operating under a fixed recovery capacity constraint, which avoids the additional instability and implementation complexity of full attacker-defender or multi-agent reinforcement learning.

The main limitation is realism. Synthetic and semi-synthetic network environments cannot capture all operational, institutional, or engineering features of real infrastructure systems. In addition, the adopted cascading-failure model abstracts from many domain-specific mechanisms that may matter in practice. The proposal therefore aims for a middle ground: enough structural realism to study recovery under constraints and cascading failure in a meaningful way, but enough abstraction to keep the learning problem tractable, interpretable, and implementable within the scope of a seminar project.

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